

Physical Chemistry (I)

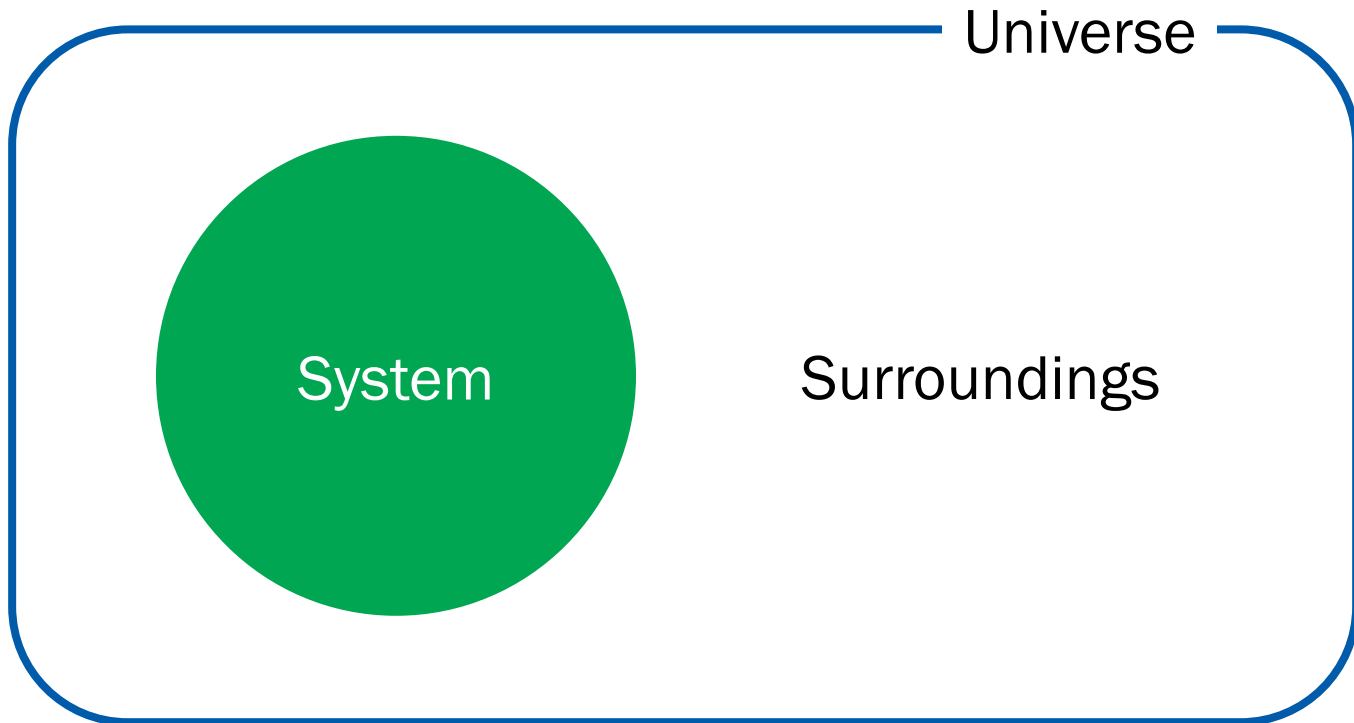
Lecture 03



In the Last Class

- What is thermodynamics?
 1. Basic Concepts
 - System: properties, equilibrium
 - System state and state change
 2. The Properties of Gases
 - Why gases?
 - Perfect gas and real gas

System and Surroundings



System Properties

- A system can have different **physical properties**
- A **single value** will be assigned for each property
- Useful classification
 - **Extensive properties:** $f(A) + f(B) = f(A + B)$
 - **Intensive properties:** $f(A) + f(B) \neq f(A + B)$

Equilibrium

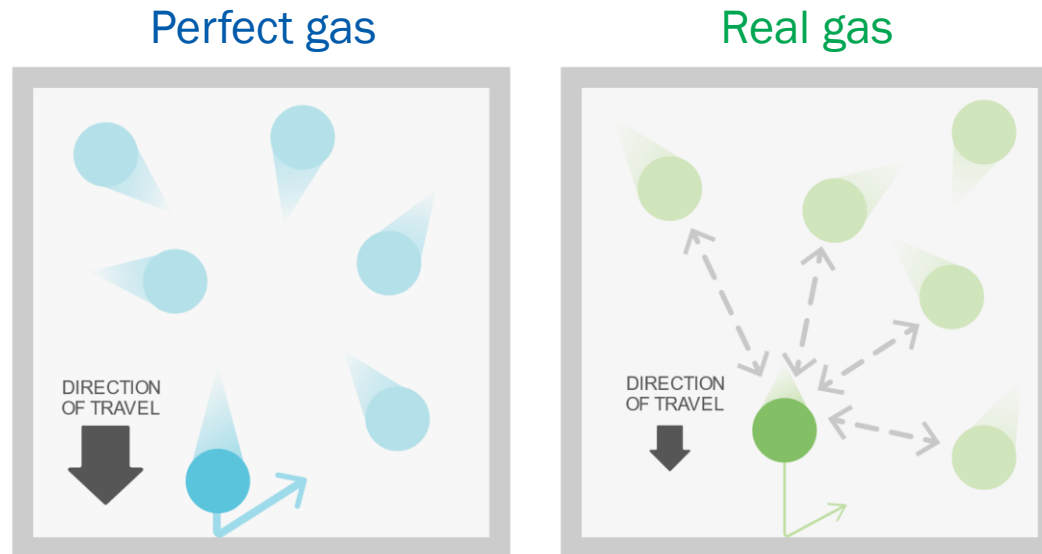
- An isolated system is **in equilibrium** when its properties remain **constant with time**
- Mechanical equilibrium: constant p and V
- Diffusive equilibrium: constant n
- Thermal equilibrium: constant T

State of a System

- The **state** of a system is defined by its **properties**
- For a pure substance: (n, V, p, T)
- **State change:** $(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$

Perfect Gas

- Perfect gas (ideal gas): no interactions
- Real gas: finite interactions between molecules



(Image: <https://cdn.kastatic.org/ka-perseus-images/4d1bf2d46543cc34f4948085bb9196f5dfb57515.svg>)

Equation of State

- The state of a pure gas is defined by (n, V, p, T)
- **Equation of state:** relationship of the 4 variables

$$p = f(T, V, n)$$

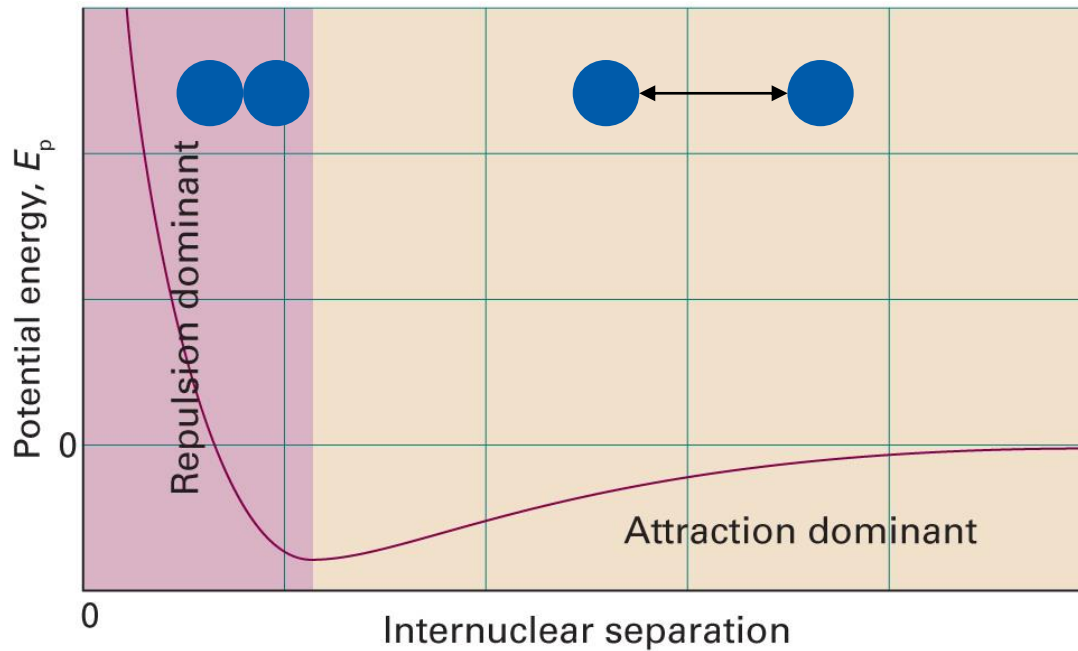
- Equation of state of a perfect gas: **perfect gas law**

$$p = nRT/V$$

↑
gas constant

Real Gas

Now the molecules interact!



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_m/V_m^0$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^0

Since $V_m^0 = RT/p$,

$$Z = \frac{pV_m}{RT} = \frac{pV}{nRT}$$

Amplitude of Z

$$Z = V_m/V_m^0$$

- $Z > 1$:

$V_m > V_m^0 \rightarrow$ repulsion is dominant

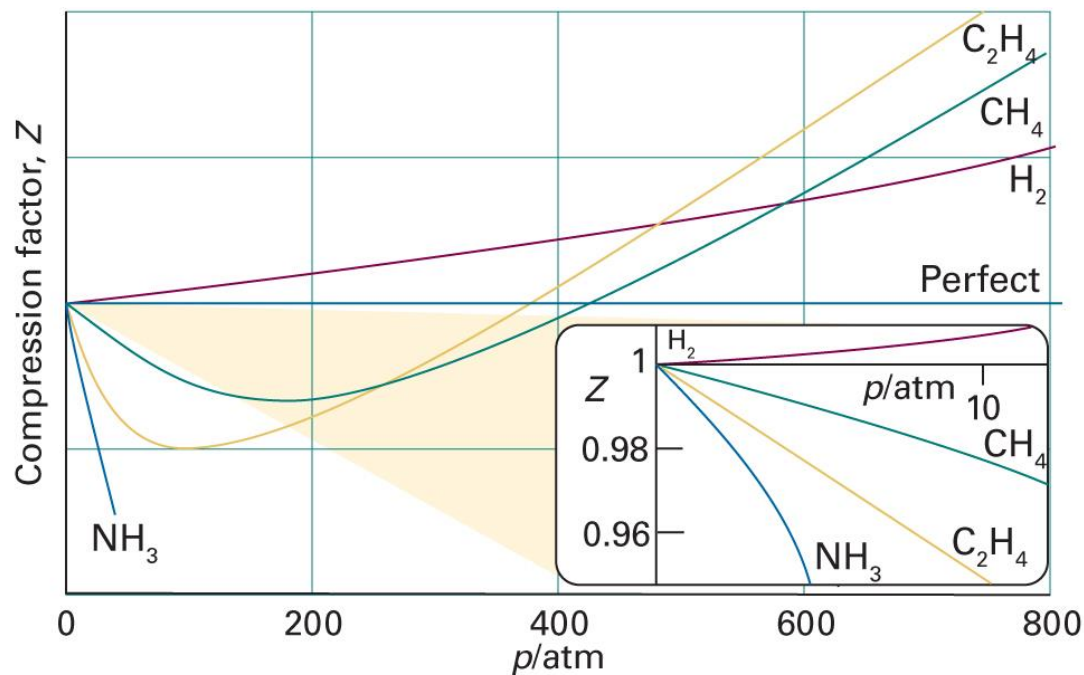
- $Z \approx 1$:

$V_m \approx V_m^0 \rightarrow$ perfect-gas-like behavior

- $Z < 1$:

$V_m < V_m^0 \rightarrow$ attraction is dominant

Experimental Values of Z



$$Z = f(p)?$$

Virial Expansion

$$Z(p) = 1 + B'p + C'p^2 + \dots$$

$$pV_m = RT(1 + B'p + C'p^2 + \dots)$$

The following expansion is more often used:

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Virial Coefficients

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

B : second virial coefficient

C : third virial coefficient

$\vdots$

- First virial coefficient = 1
- Virial coefficients are temperature-dependent

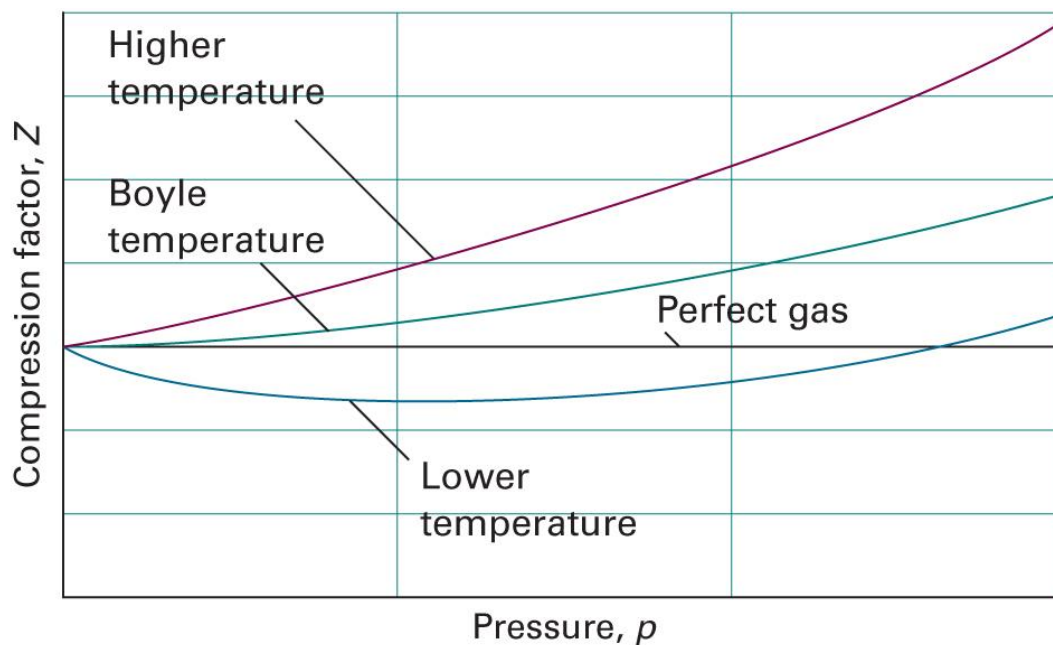
Second Virial Coefficient

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

- Generally, $B/V_m \gg C/V_m^2$
- $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
- $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
- $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

Boyle Temperature T_B

Temperature where $B = 0$



Experimental Data

	$B / (\text{cm}^3 \text{ mol}^{-1})$		T_B / K
	$T = 273 \text{ K}$	$T = 600 \text{ K}$	
Ar	-21.7	11.9	411.5
CO ₂	-149.7	-12.4	714.8
N ₂	-10.5	21.7	327.2
Xe	-153.7	-19.6	768.0

Brief Illustration 1C.2

Use the virial expansion (up to the B term) to calculate the pressure exerted at 100 K by 0.104 mol $\text{O}_2(\text{g})$ in a vessel of volume 0.225 L.

$$pV_m = RT \left(1 + \frac{B}{V_m} \right)$$

$$V_m = V/n = (0.225 \text{ L}) / (0.104 \text{ mol}) = 2.16 \text{ L mol}^{-1}$$

$$\text{For } \text{O}_2(\text{g}), B = -197.5 \text{ cm}^3 \text{ mol}^{-1} = -0.1975 \text{ L mol}^{-1}$$

Now, substitute the values into

$$p = \frac{RT}{V_m} \left(1 + \frac{B}{V_m} \right)$$

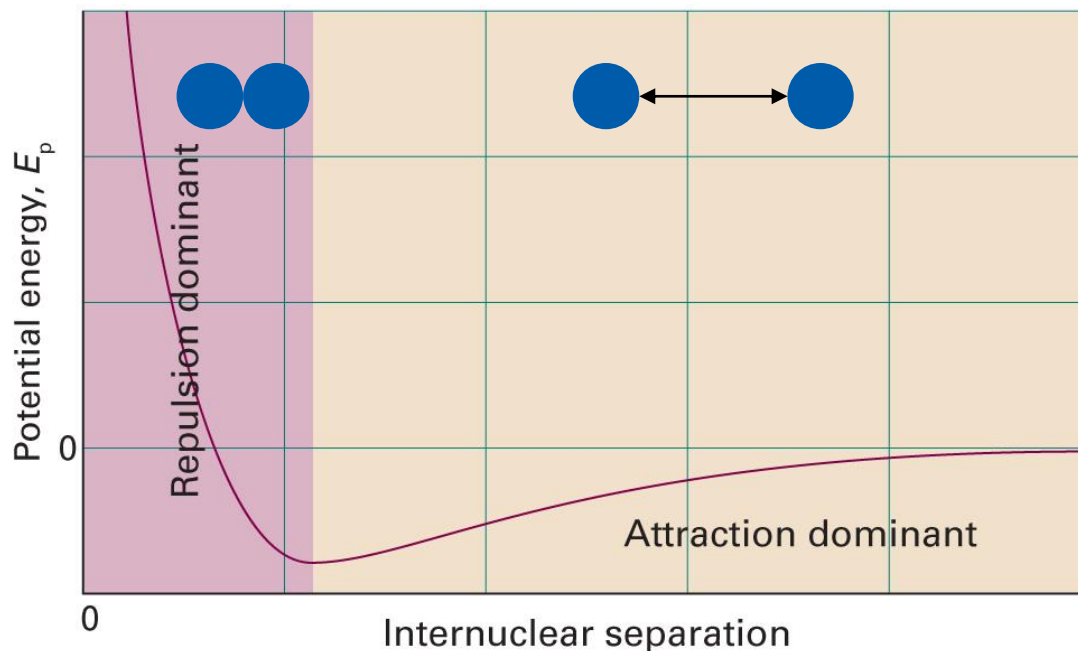
Brief Illustration 1C.2

$$V_m = 2.16 \text{ L mol}^{-1}, B = -0.1975 \text{ L mol}^{-1}, T = 100 \text{ K}$$

$$\begin{aligned} p &= \frac{RT}{V_m} \left(1 + \frac{B}{V_m} \right) \\ &= \frac{(0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1})(100 \text{ K})}{2.16 \text{ L mol}^{-1}} \left(1 - \frac{0.1975 \text{ L mol}^{-1}}{2.16 \text{ L mol}^{-1}} \right) \\ &= 3.45 \text{ atm} \end{aligned}$$

Corrected Perfect Gas Law

Consider the effects of **repulsion** and **attraction**



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

van der Waals Equation

- **Repulsion:** decrease of the total volume

$$V \rightarrow V - nb$$

$$p = \frac{nRT}{V - nb}$$

- **Attraction:** decrease of the pressure

$p \sim$ (frequency of collisions) $\times$ (force of each particle)

decreases as $n/V \uparrow$

decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals Coefficients

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

- a : strength of attractive interactions
- b : strength of repulsive interactions
- Characteristics of each gas
- Independent of the temperature

Experimental Data

	$a / (\text{atm L}^2 \text{ mol}^{-2})$	$b / (10^{-2} \text{ L mol}^{-1})$
Ar	1.337	3.20
CO ₂	3.610	4.29
He	0.0341	2.38
Xe	4.137	5.16

Equation Manipulations

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$V_m^3 - \left(b + \frac{RT}{p}\right)V_m^2 + \left(\frac{a}{p}\right)V_m - \frac{ab}{p} = 0$$

→ cubic equation

Example 1C.1

Estimate the molar volume of CO_2 at 500 K and 100 atm by treating it as a van der Waals gas.

Parameters for CO_2 : $a = 3.610 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 4.29 \times 10^{-2} \text{ L mol}^{-1}$

$$V_m^3 - \left(b + \frac{RT}{p}\right) V_m^2 + \left(\frac{a}{p}\right) V_m - \frac{ab}{p} = 0$$

Coefficients are

$$b + RT/p = 4.29 \times 10^{-2} \text{ L mol}^{-1} + (0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}) (500 \text{ K}) / (100 \text{ atm}) = 0.453 \text{ L mol}^{-1}$$

$$a/p = (3.610 \text{ atm L}^2 \text{ mol}^{-2}) / (100 \text{ atm}) = 3.61 \times 10^{-2} \text{ L}^2 \text{ mol}^{-2}$$

$$ab/p = (3.610 \text{ atm L}^2 \text{ mol}^{-2}) (4.29 \times 10^{-2} \text{ L mol}^{-1}) / (100 \text{ atm}) = 1.55 \times 10^{-3} \text{ L}^3 \text{ mol}^{-3}$$

Example 1C.1

Let $x = V_m/(\text{L mol}^{-1})$. Then

$$x^3 - 0.453x^2 + (3.61 \times 10^{-2})x - 1.55 \times 10^{-3} = 0$$

A numerical solver gives

$$x = 0.366$$

Thus, $V_m = 0.366 \text{ L mol}^{-1}$.

van der Waals and Virial

- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Virial expansion

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Can we connect the two?

Some Math

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$\text{vdW: } p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$p = \frac{1}{V_m} \left(\frac{RT}{1 - b/V_m} \right) - \frac{a}{V_m^2}$$

$$pV_m = \frac{RT}{1 - b/V_m} - \frac{a}{V_m}$$

$$pV_m = RT \left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m} \right)$$

Some More Math

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$pV_m = RT \left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m} \right)$$

As $b/V_m \ll 1$, we use

$$\frac{1}{1 - x} = 1 + x + x^2 + \dots \text{ (when } x < 1 \text{)}$$

$$pV_m = RT \left[\left\{ 1 + \frac{b}{V_m} + \left(\frac{b}{V_m} \right)^2 + \dots \right\} - \frac{a/RT}{V_m} \right]$$

Finally

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$pV_m = RT \left[\left\{ 1 + \frac{b}{V_m} + \left(\frac{b}{V_m} \right)^2 + \dots \right\} - \frac{a/RT}{V_m} \right]$$

$$pV_m = RT \left[1 + \left(b - \frac{a}{RT} \right) \frac{1}{V_m} + \frac{b^2}{V_m^2} + \dots \right]$$

Hence,

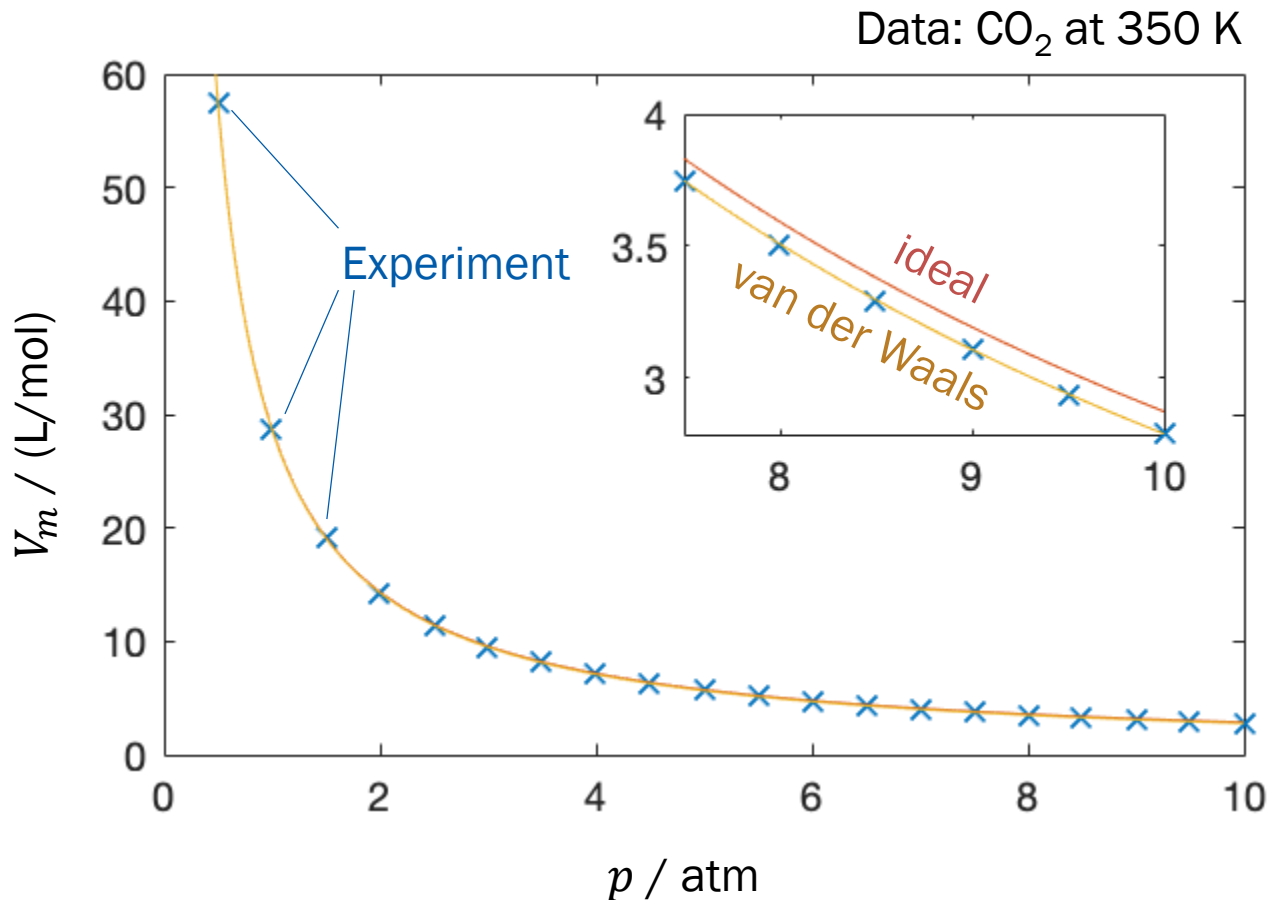
$$B = b - a/RT \text{ and } C = b^2$$



Also, from $B = b - a/RT = 0$, $T_B = a/(bR)$

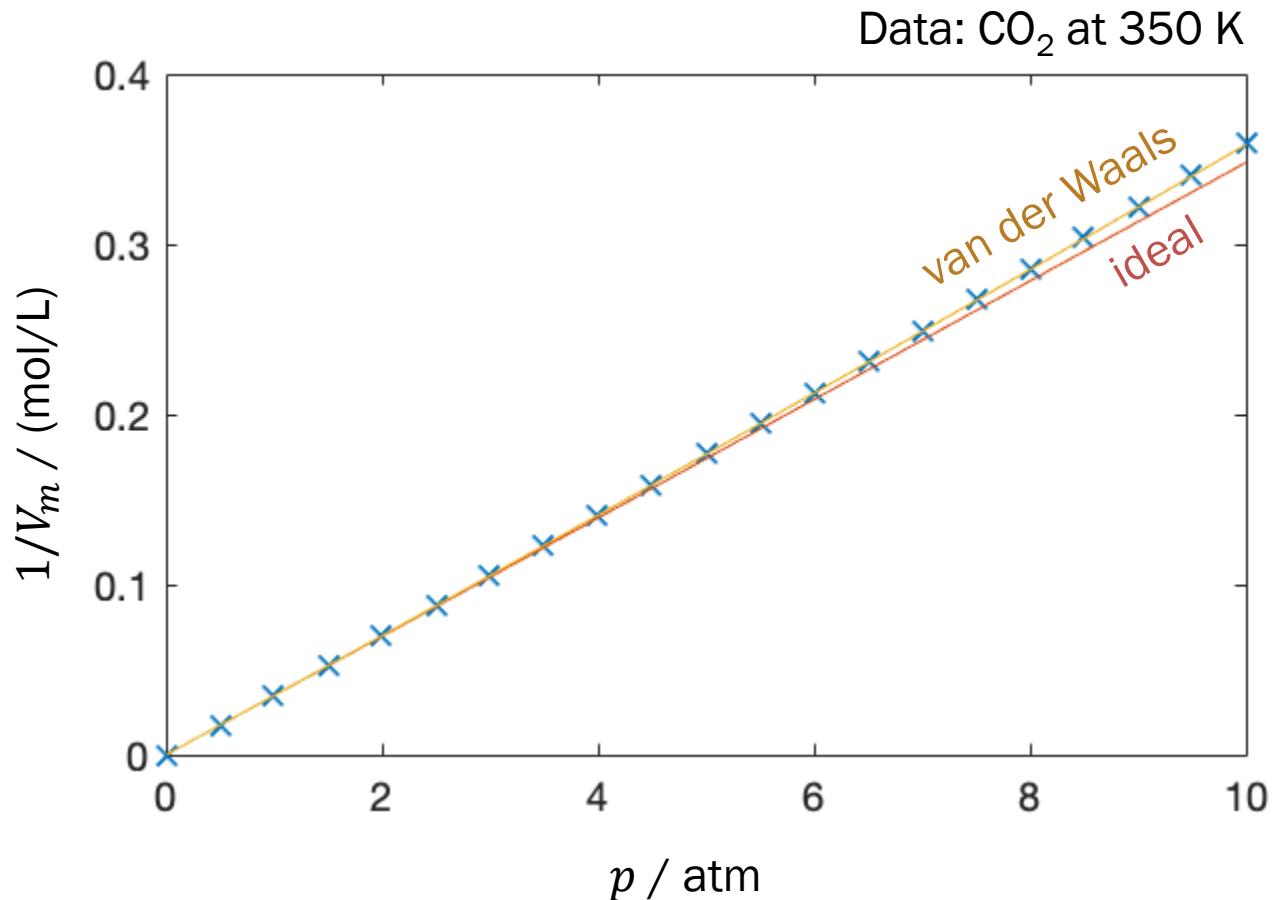
Not in Atkins

Validity of van der Waals



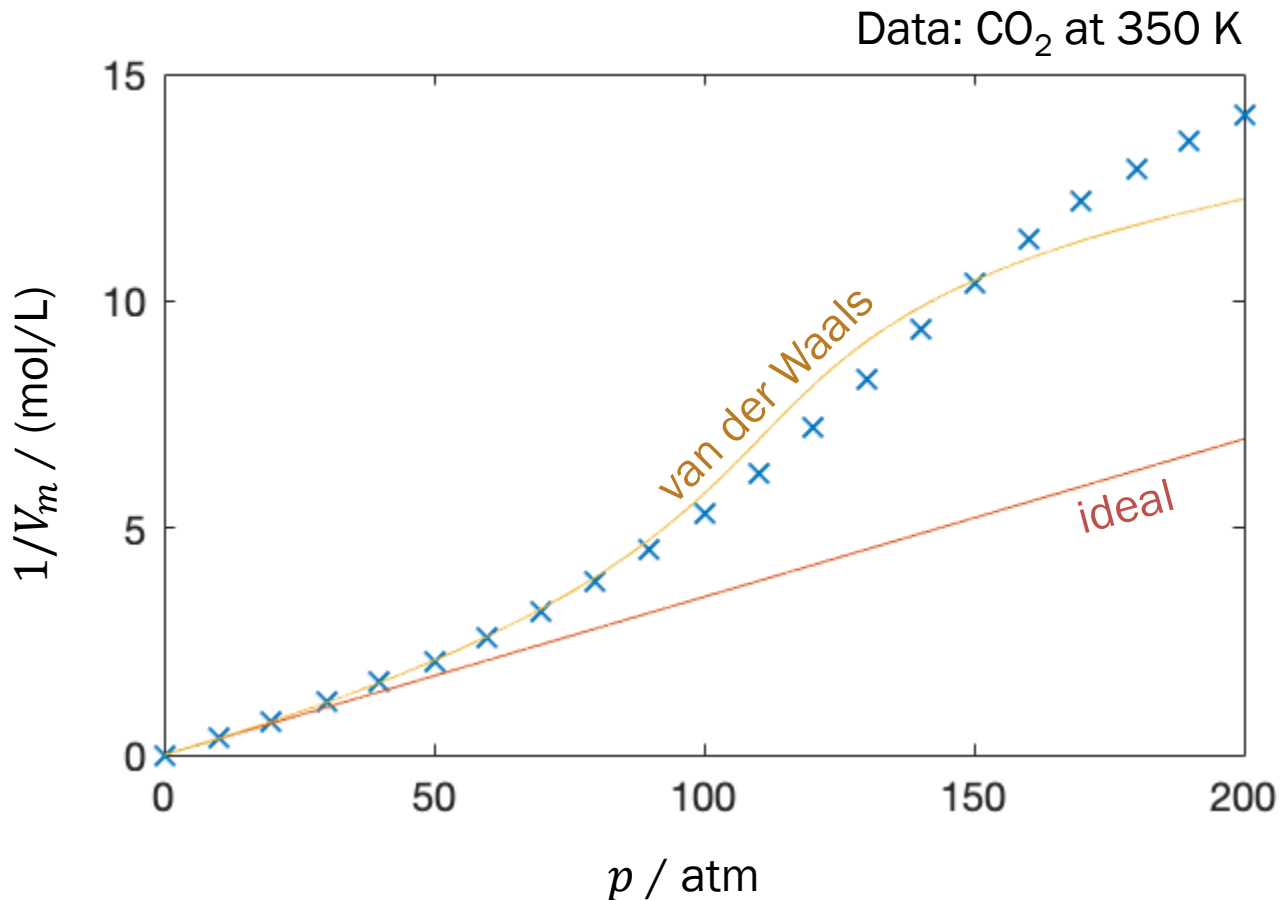
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Validity of van der Waals



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Is van der Waals Perfect?



Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the attraction term
 - Redlich-Kwong equation (1949)

$$p = \frac{RT}{V_m - B_{RK}} - \frac{A_{RK}}{V_m(V_m + B_{RK})T^{1/2}}$$

- Peng-Robinson equation (1976)

$$p = \frac{RT}{V_m - B_{PR}} - \frac{A_{PR}}{V_m(V_m + B_{PR}) + B_{PR}(V_m - B_{PR})}$$

Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the repulsion term
 - Guggenheim equation (1965)

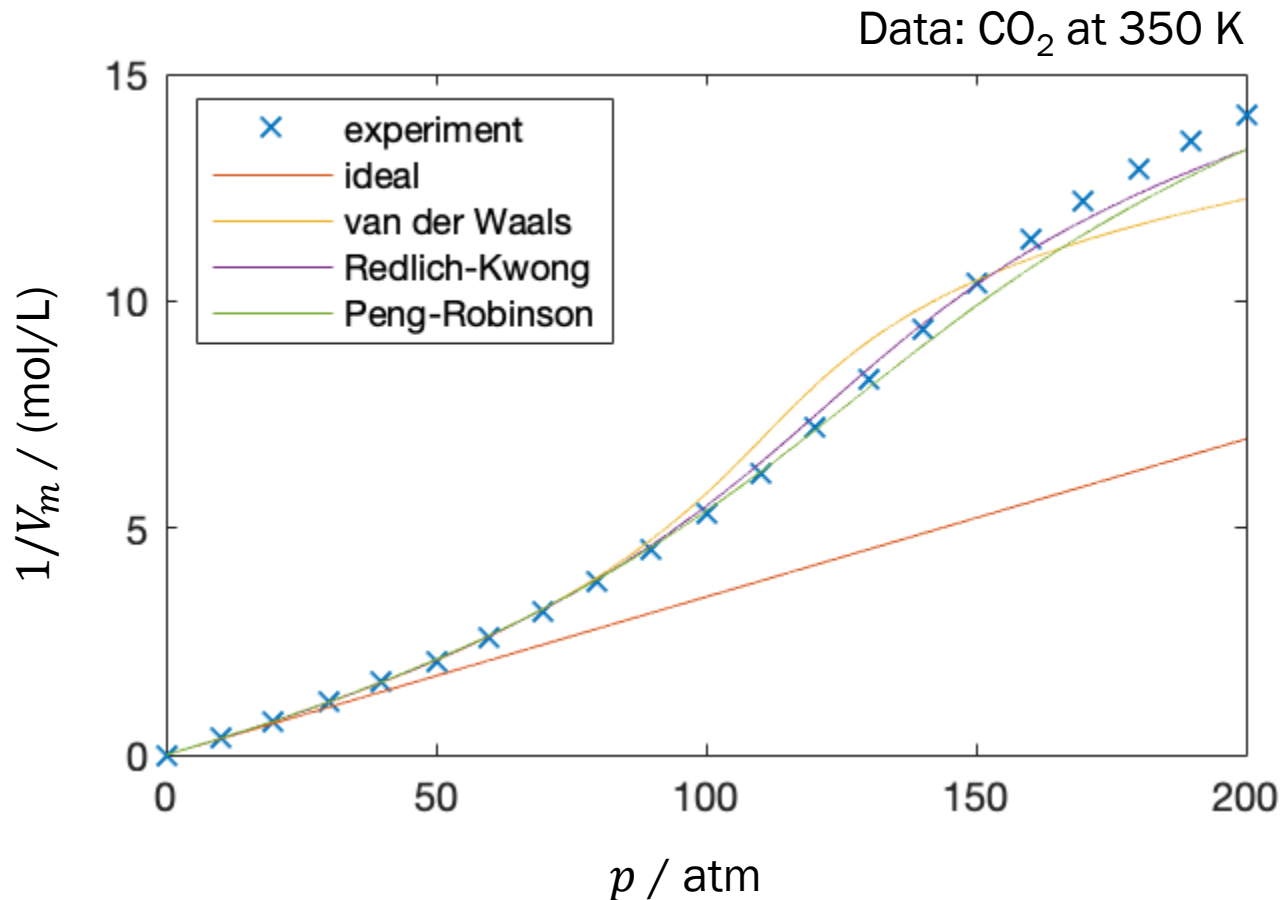
$$p = \frac{RT}{V_m(1 - y)^4} - \frac{A_G}{V_m^2}, \quad y = \frac{B_G}{V_m}$$

- Carnahan-Starling-van der Waals equation (1972)

$$p = \frac{RT(1 + y + y^2 - y^3)}{V_m(1 - y)^3} - \frac{A_{CSW}}{V_m^2}, \quad y = \frac{B_{CSW}}{V_m}$$

Not in Atkins

Better Descriptions



Now we are going back to the perfect gas!

Kinetic Molecular Theory

Total energy = Σ (kinetic energies of molecules)

Speed distribution

Collision frequency

Pressure

Internal Energy

⋮

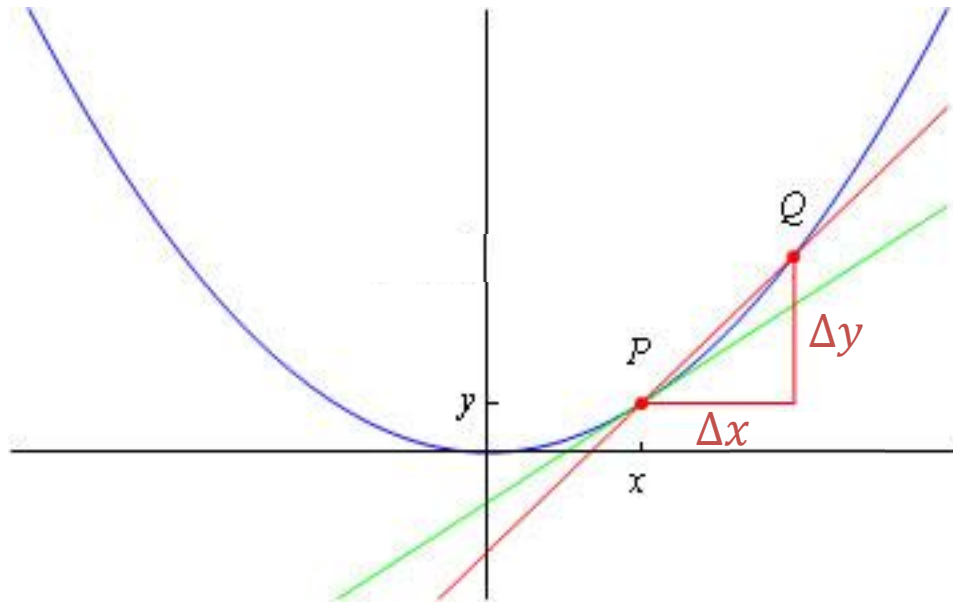
Fundamental Assumptions

1. Molecules with mass m in ceaseless random motion
2. Molecules with negligible size
3. Elastic collisions: the total kinetic energy is conserved

Described by classical mechanics

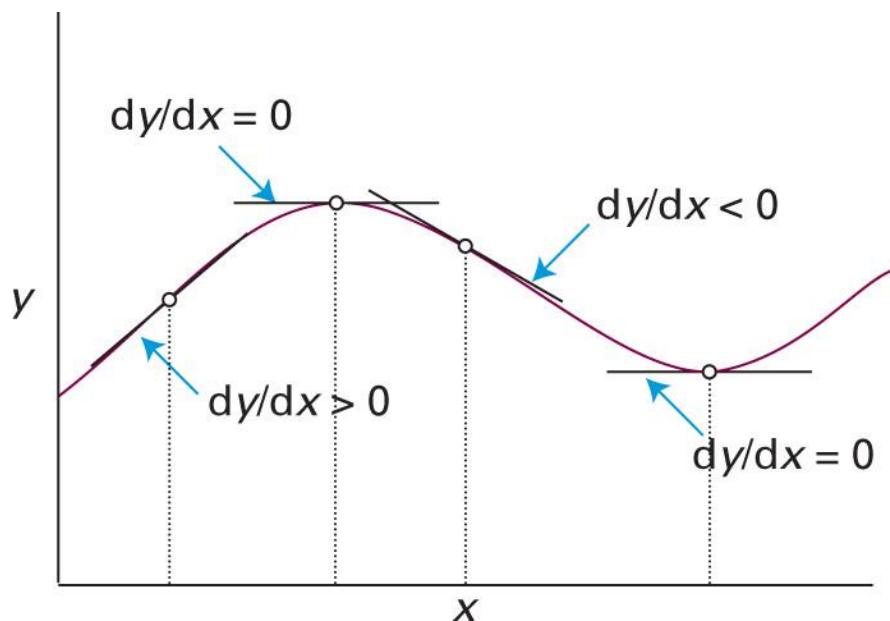
Differentiation

“The slope of a function”



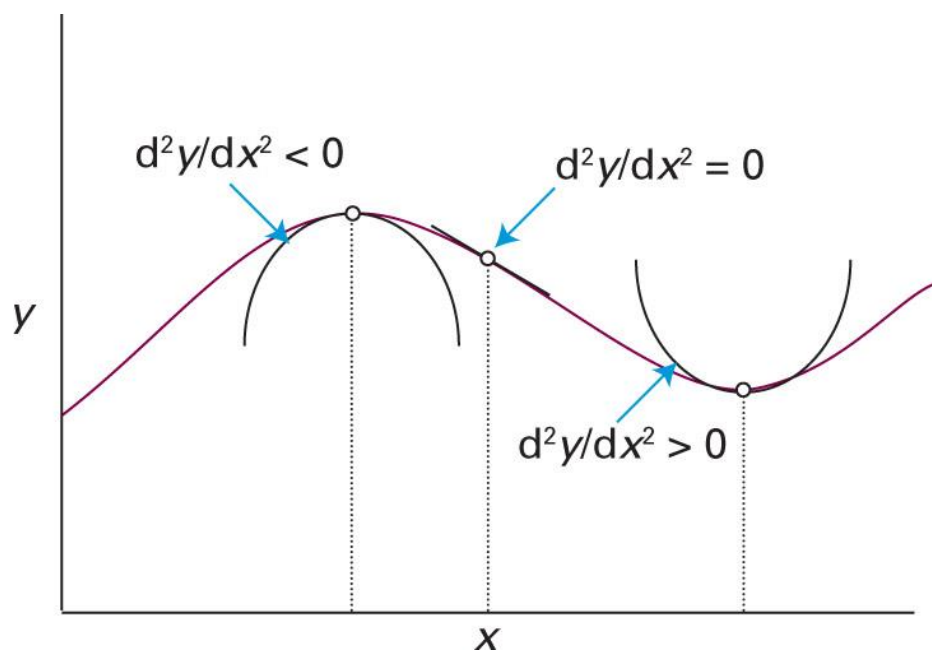
$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

First Derivative: Slope



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., the chemist's toolkit 5, Sketch 1)

Second Derivative: Sharpness



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., the chemist's toolkit 5, Sketch 2)

Derivatives of Some Functions

$$\frac{d}{dx} a = 0$$

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^{ax} = ae^{ax}$$

$$\frac{d}{dx} \ln ax = \frac{1}{x}$$

$$\frac{d}{dx} \sin ax = a \cos ax$$

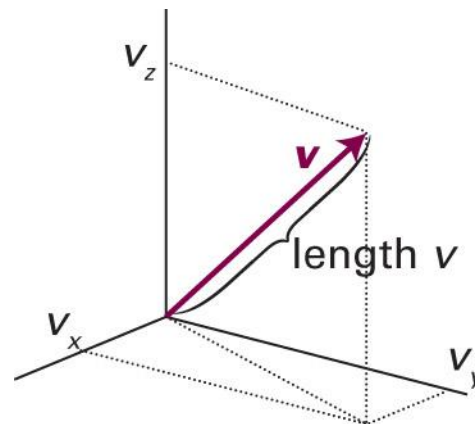
$$\frac{d}{dx} \cos ax = -a \sin ax$$

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

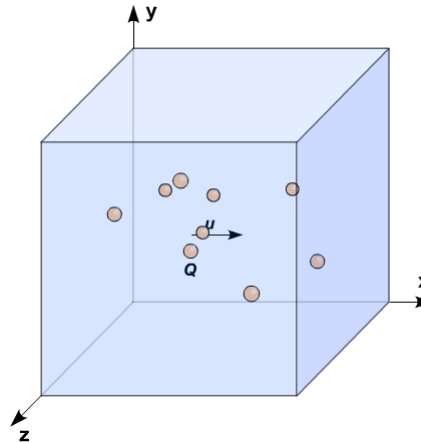


With the velocity, we can define:

- Speed $v = |\vec{v}| = (v_x^2 + v_y^2 + v_z^2)^{1/2}$
- Kinetic energy $E_k = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Molecular Velocities

Each molecule has a different velocity



How can we determine the information about velocity?